

Exercise - Week 3: MATH1061/7861

Duration: 30 minutes

3 Questions - 10 marks

Topics: Conditional Statements, Valid and Invalid Arguments, Methods for Determining Validity, and Quantified Statements and their Negations

Student Name:

Student number: |_|_|_|_|_|_|_|_|

Your student number should be 8 digits and **not** start with an s. e.g. 41234567

- (1) (**3 marks**) Is the following statement true or false? Justify your answer.

$$(p \wedge q) \rightarrow r \equiv (\sim p) \rightarrow (q \rightarrow r).$$

- (2) (**3 marks**) Is the following argument valid? Justify your answer.

$$p \rightarrow q$$

$$p \vee r$$

$$p \vee \sim r$$

$$\therefore \sim q$$

- (3) **(4 marks)** For each of the following statements, write the negation. Determine whether the original statement or the negation is true. Justify your answer.

(a) $\forall x \in \mathbb{R}, (x^2 \geq x) \rightarrow (x \geq 0)$.

(b) $\forall x, y \in \mathbb{R}$, if $x + y \in \mathbb{Z}$ and $x - y \in \mathbb{Z}$ then $x \in \mathbb{Z}$.